

THIRUMALAI SUBASHREE

Electronics and Communication Engineering Student | Embedded Systems & IoT

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PROFILE

Third-year Electronics and Communication Engineering student with hands-on experience building embedded hardware-software systems using microcontrollers and IoT technologies. Skilled in developing complete prototypes spanning firmware development, sensor and communication interfacing across multiple microcontroller platforms, including a cloud and mobile-connected IoT system built on ESP32-S3. Seeking an internship opportunity in Embedded Systems or IoT.

EDUCATION

Vellore Institute of Technology, Chennai

2024 - 2028 (Expected)

B.Tech, Electronics and Communication Engineering | CGPA: 9.31/10

Velammal Bodhi Campus, Kolapakkam

Senior Secondary Examination (Class XII) | Score: 89.9%

SKILLS SUMMARY

- **Microcontrollers:** ESP32, ESP32-S3, STM32, Arduino, Raspberry Pi Pico
- **Communication Protocols:** WiFi, HTTP, I2C
- **Languages:** C, Embedded C, Java, Python
- **Tools:** STM32CubeIDE, STM32CubeMX, Keil µVision, MATLAB, Git
- **Simulation & Digital Design:** Cadence Virtuoso, Sentaurus TCAD, LTSpice, Quartus, ModelSim
- **Cloud, Backend & Mobile:** Firebase, Node.js, Flutter

PROJECTS

SheAlert - Women's Safety Monitoring System

ESP32-S3, Flutter, Firebase, Node.js

- Engineered an end-to-end embedded IoT system pairing an ESP32-S3 device with a Flutter mobile app, Firebase, and a Node.js backend; automatic mode detects a trigger phrase and captures photo/audio evidence, then alerts emergency contacts via WhatsApp with live location
- Designed a manual 2-second-hold SOS mode as a fast, low-friction backup trigger for emergency alerts

Smart Coffee Maker

8052 Microcontroller, Embedded C

- Developed an automated coffee-dispensing embedded system featuring IR-based cup detection, dual motor control, and an I2C LCD status interface

Combinational Lock with Alert Mechanism

D Flip-Flops, Comparators, Counters

- Designed a secure 4-bit digital lock using pure combinational and sequential logic (no microcontroller); D flip-flops store the password, comparators verify entries, and a counter triggers a red-LED alert after repeated failed attempts

Gas Alert - Intelligent Sensor Detection System

Arduino UNO, MQ9 Sensor, I2C LCD

- Built a standalone gas-leak detection system that displays live sensor readings on an I2C LCD and triggers buzzer/LED alerts when readings cross a calibrated safety threshold

14nm Double-Gate MOSFET Amplifier Design

Sentaurus TCAD, Verilog-A, Cadence Virtuoso

- Modeled a 14nm double-gate MOSFET in TCAD, extracted a Verilog-A compact model from it, then used the model to design and analyze a Common Source Amplifier in Cadence Virtuoso

WORK EXPERIENCE

Embedded Systems Intern | Evolve Robot Lab

Jun 2026 - Jul 2026

- Integrated IoT connectivity on ESP32-S3 with ThingSpeak and Blynk for real-time monitoring and control
- Prototyped with STM32 and Raspberry Pi Pico for LED control and sensor interfacing, and worked with ESP32-S3 audio/camera modules, laying the groundwork for the SheAlert project
- Built Arduino-based robotics projects, including a robotic arm and an obstacle-avoiding robot, covering sensor interfacing and motor control
- Simulated and validated circuit designs in Tinkercad before hardware implementation, reducing wiring errors during build

CERTIFICATIONS & COURSES

- HackerRank Certification - Python (Basic)
- Simplilearn: VLSI Design, Embedded Systems, IoT, PCB Design